

Real Numbers (9 th std)

Choose the Appropriate Answer

1. If n is a natural number then n is

- (a) Always a natural number. (b) Always an irrational number.
(b) Always a rational number (d) May be rational or irrational

Ans: (d) May be rational or irrational

2. Which of the following is not true?

- (a) Every rational number is a real number.
(b) Every integer is a rational number.
(c) Every real number is an irrational number.
(d) Every natural number is a whole number.

Ans: (c) Every real number is an irrational number.

3. Which one of the following, regarding sum of two irrational numbers, is true?

- (a) Always an irrational number. (b) May be a rational or an irrational number.
(c) Always a rational number. (d) Always an integer.

Ans: (b) May be a rational or an irrational number.

4. Which one of the following has a terminating decimal expansion?

- (a) $5/64$ (b) $8/9$ (c) $14/15$ (c) $1/12$

Ans: (a) $5/64$

5. Which one of the following is an irrational number

- (a) $\sqrt{25}$ (b) $\sqrt{9/4}$ (c) $7/11$ (d) π

Ans: (d) π

6. An irrational number between 2 and 2.5 is

- (a) $\sqrt{11}$ (b) $\sqrt{5}$ (c) $\sqrt{2.5}$ (d) $\sqrt{8}$

Ans: (b) $\sqrt{5}$

7. The smallest rational number by which $1/3$ should be multiplied so that its decimal expansion terminates with one place of decimal is

- (a) $1/10$ (b) $3/10$ (c) 3 (d) 30

Ans: (b) $3/10$

8. Find the odd one out of the following.

- (a) $\sqrt{32} \times \sqrt{2}$ (b) $\sqrt{27} / \sqrt{3}$ (c) $\sqrt{72} \times \sqrt{8}$ (d) $\sqrt{54} / \sqrt{8}$

Ans: (b) $\sqrt{27} / \sqrt{3}$

9. $0.343434 + 0.343434 =$

- (a) 0.68787 (b) 0.6868 (c) 0.68888 (d) 0.687777

Ans: (a) 0.68787

10. Which of the following statement is false?

- (a) The square root of 25 is 5 or -5 (b) $\sqrt{25} = 5$
(c) $-\sqrt{25} = -5$ (d) $\sqrt{25} = \pm 5$

Ans: (d) $\sqrt{25} = \pm 5$

11. Which one of the following is not a rational number?

- (a) $\sqrt{8/18}$ (b) $7/3$ (c) $\sqrt{0.01}$ (d) $\sqrt{13}$

Ans: (d) $\sqrt{13}$

12. $\sqrt{27} + \sqrt{12} =$

- (a) $\sqrt{39}$ (b) $5\sqrt{6}$ (c) $5\sqrt{3}$ (d) $3\sqrt{5}$

Ans: (d) $3\sqrt{5}$

13. If $\sqrt{80} = k\sqrt{5}$, then

- (a) 2 (b) 4 (c) 8 (d) 16

Ans: (b) 4

14. $4\sqrt{7} \times 2\sqrt{3} =$

- (a) $6\sqrt{10}$ (b) $8\sqrt{21}$ (c) $8\sqrt{10}$ (d) $6\sqrt{21}$

Ans: (b) $8\sqrt{21}$

15. When written with a rational denominator, the expression $2\sqrt{3}/3\sqrt{2}$ can be simplified as

- (a) $\sqrt{2/3}$ (b) $\sqrt{3/2}$ (c) $\sqrt{6/3}$ (d) $2/3$

Ans: (c) $\sqrt{6/3}$

16. When $(2\sqrt{5} - \sqrt{2})^2$ is simplified, we get

- (a) $4\sqrt{5} + 2\sqrt{2}$ (b) $22 - 4\sqrt{10}$ (c) $8 - 4\sqrt{10}$ (d) $2\sqrt{10} - 2$

Ans: (b) $22 - 4\sqrt{10}$

17. $(0.000729)^{-3/4} \times (0.09)^{-3/4}$

- (a) $10^3/3^3$ (b) $10^5/3^5$ (c) $10^2/3^2$ (d) $10^6/3^6$

Ans: (d) $10^6/3^6$

18. If $\sqrt[3]{9^x} = \sqrt[3]{9^2}$, then $x =$ _____

- (a) $2/3$ (b) $4/3$ (c) $1/3$ (d) $5/3$

Ans: (b) $4/3$

19. The length and breadth of a rectangular plot are 5×10^5 and 4×10^4 meters respectively. Its area is _____.

- (a) $9 \times 10^1 \text{ m}^2$ (b) $9 \times 10^9 \text{ m}^2$ (c) $2 \times 10^{10} \text{ m}^2$ (d) $20 \times 10^{20} \text{ m}^2$

Ans: (c) $2 \times 10^{10} \text{ m}^2$

20. On simplifying $16^3 \times 2^4$, we get

- (a) 2^{15} (b) 2^{16} (c) 2^{10} (d) 2^{12}

Ans: (b) 2^{16}

$$16^3 \times 2^4 \Rightarrow (2^4)^3 \times 2^4 \Rightarrow 2^{12} \times 2^4 = 2^{12+4} = 2^{16}$$

21. Can we write 0 in the form of p/q ?

- (a) Yes (b) No (c) Can't be explained (d) None of the above

Ans: (a) Yes

22. The three rational numbers between 3 and 4 are:
 (a) $5/2, 6/2, 7/2$ (b) $13/4, 14/4, 15/4$
 (c) $12/7, 13/7, 14/7$ (d) $11/4, 12/4, 13/4$
Ans: (b) $13/4, 14/4, 15/4$

23. In between any two numbers, there are:
 (a) Only one rational number (b) Two rational numbers
 (c) Infinite rational numbers (d) No rational number
Ans: (c) Infinite rational numbers

24. Every rational number is _____ number.
 (a) Whole (b) Natural (c) Integer (d) Real
Ans: (d) Real

25. $\sqrt{9}$ is _____ number.
 (a) A rational (b) An irrational
 (c) Neither rational nor irrational (d) None of the above
Ans: (a) A rational

26. Which of the following is an irrational number?
 (a) $\sqrt{16}$ (b) $\sqrt{12/3}$ (c) $\sqrt{12}$ (d) $\sqrt{100}$
Ans: (c) $\sqrt{12}$

27. $3\sqrt{6} + 4\sqrt{6}$ is equal to:
 (a) $6\sqrt{6}$ (b) $7\sqrt{6}$ (c) $4\sqrt{12}$ (d) $7\sqrt{12}$
Ans: (b) $7\sqrt{6}$

28. $\sqrt{6} \times \sqrt{27}$ is equal to:
 (a) $9\sqrt{2}$ (b) $3\sqrt{3}$ (c) $2\sqrt{2}$ (d) $9\sqrt{3}$
Ans: (a) $9\sqrt{2}$

29. Which of the following is equal to x^3 ?
 (a) $x^6 - x^3$ (b) $x^6 \cdot x^3$ (c) x^6 / x^3 (d) $(x^6)^3$
Ans: (c) x^6 / x^3

30. Which of the following is an irrational number?
 (a) $\sqrt{23}$ (b) $\sqrt{225}$ (c) 0.3796 (d) 7.478478
Ans: (a) $\sqrt{23}$

31. Which of the following is an irrational number?
 (a) 0.14 (b) 0.14161616... (c) 0.14161416... (d)
 0.4014001400014...
Ans: (d) 0.4014001400014...

32. $2\sqrt{3} + \sqrt{3} =$
 (a) 6 (b) $2\sqrt{6}$ (c) $3\sqrt{3}$ (d) $4\sqrt{6}$
Ans: (c) $3\sqrt{3}$

33. The number obtained on rationalising the denominator of $1 / (\sqrt{7} - 2)$ is
(a) $(\sqrt{7}+2) / 3$ (b) $(\sqrt{7}-2) / 3$ (c) $(\sqrt{7}+2) / 5$ (d) $(\sqrt{7}+2) / 45$

Ans: (a) $(\sqrt{7}+2) / 3$

34. Which of the following is rational?

(a) $4 / 0$ (b) $0 / 4$ (c) $\sqrt{3}$ (d) π

Ans: (b) $0 / 4$

35. The irrational number between 2 and 2.5 is

(a) $\sqrt{11}$ (b) $\sqrt{5}$ (c) $\sqrt{22.5}$ (d) $\sqrt{12.5}$

Ans: (b) $\sqrt{5}$

36. The value of $\sqrt{10}$ times $\sqrt{15}$ is equal to

(a) $5\sqrt{6}$ (b) $\sqrt{25}$ (c) $10\sqrt{5}$ (d) $\sqrt{5}$

Ans: (a) $5\sqrt{6}$

37. The decimal representation of the rational number is

(a) Always terminating (b) Either terminating or repeating
(c) Either terminating or non-repeating (d) Neither terminating nor repeating

Ans: (b) Either terminating or repeating

38. Which of the following is a rational number?

(a) 0 (b) $2\sqrt{3}$ (c) $2+\sqrt{3}$ (d) π

Ans: (a) 0

39. Which of the following is an irrational number?

(a) $\sqrt{4 / 9}$ (b) $\sqrt{12} / \sqrt{3}$ (c) $\sqrt{7}$ (d) $\sqrt{81}$

Ans: (c) $\sqrt{7}$

40. The product of a rational and an irrational number is:

(a) Always rational (b) Always irrational
(c) Sometimes rational, sometimes irrational (d) Always an integer

Ans: (b) Always irrational

41. Every integer is a / an _____ number.

(a) Natural (b) Whole (c) Rational (d) Irrational

Ans: (c) Rational

42. Which of the following is Natural number?

(a) 3.5 (b) 7 (c) 0 (d) -2

Ans: (b) 7

43. Integer numbers set is denoted by which letter?

(a) N (b) W (c) Q (d) Z

Ans: (d) Z

44. Zero is a rational number.

(a) False (b) True (c) Can't determine (d) None

Ans: (b) True

45. Find the incorrect option (Assume conventional notations).

- (a) $Q = p / q$ where p and q are integers and $q \neq 0$ (b) $N = \{1, 2, 3, \dots\}$
(c) $W = \{0, 1, 2, 3, \dots\}$ (d) $Z = \{\dots -2, -1, 0, 1, 2, \dots\}$

Ans: (a) $Q = p / q$ where p and q are integers and $q \neq 0$

46. There are _____ rational numbers between 2 and 3.

- (a) Zero (b) Two (c) Infinite (d) Data is insufficient

Ans: (c) Infinite

47. Which of the following option is true based on the below statements?

- (i) All integers are rational numbers.
(ii) All rational numbers are integers.

- (a) Both statements are true
(b) Statement (i) is true and Statement (ii) is false
(c) Statement (i) is false and Statement (ii) is true
(d) Both statements are false

Ans: (b) Statement (i) is true and Statement (ii) is false

48. Which of the following is not rational number?

- (a) $8 / 9$ (b) $5 / 7$ (c) $-3 / 2$ (d) $\sqrt{2}$

Ans: (d) $\sqrt{2}$

49. How many whole numbers are there between -3 and 3 (Including -3 and 3)?

- (a) Four (b) Zero (c) Seven (d) Three

Ans: (a) Four

50. N is subset of W , Z and Q . (Assume conventional notations)

- (a) True (b) False (c) Can't determine (d) None

Ans: (a) True

51. Rational numbers are denoted by _____

- (a) N (b) Z (c) Q (d) W

Ans: (c) Q

52. Which of the following is the correct way to write $5.412\overline{}$?

- (a) $5.412412412\dots$ (b) $5.4121212\dots$ (c) -5.412 (d) $5.41212\dots$

Ans: (b) $5.4121212\dots$

53. $2.03454545\dots =$ _____

- (a) $345 / 1000$ (b) $350 / 995$ (c) $355 / 900$ (d) $342 / 990$

Ans: (d) $342 / 990$

54. The decimal expansion of rational numbers is either terminating or non-terminating and recurring (repeating).

- (a) True (b) False (c) Can't determine (d) None

Ans: (a) True

55. $0.3333\ldots$ is a / an _____ number.
(a) Natural (b) Whole (c) Rational (d) Irrational
Ans: (c) Rational

56. $0.238765563246\ldots$ is a / an _____ number.
(a) Rational (b) Irrational (c) Natural (d) Whole
Ans: (b) Irrational

57. Which of the following number is an irrational number between $1/3$ and $2/3$?
(a) $0.3333\ldots$ (b) $0.343403400\ldots$ (c) $0.5555\ldots$ (d) $0.67670671\ldots$
Ans: (b) 0.343403400...

58. There are infinite irrational numbers between two numbers.
(a) True (b) False (c) Can't Determine (d) None
Ans: (a) True

59. $\sqrt{5}$ is a / an _____ number.
(a) Natural (b) Whole (c) Rational (d) Irrational
Ans: (d) Irrational

60. Summation or Subtraction of two non-zero rational numbers is a / an _____.
(a) Natural number (b) Whole number
(c) Rational Number (d) Irrational number
Ans: (c) Rational Number

61. If we multiply or divide two rational numbers, we get a / an _____ number.
(a) Natural (b) Whole (c) Rational (d) Irrational
Ans: (c) Rational

62. If we add or subtract an irrational number and a rational number (non-zero), then we get a / an _____ number.
(a) Natural (b) Rational (c) Irrational (d) Rational or Irrational
Ans: (c) Irrational

63. If we add or subtract two irrational numbers, we get _____ number.
(a) Natural (b) Rational (c) Irrational (d) Rational or Irrational
Ans: (d) Rational or Irrational

64. $3 + \sqrt{-2}$ is a / an _____ number.
(a) Natural (b) Irrational (c) Rational (d) Whole
Ans: (b) Irrational

65. $\pi - \pi$ is _____ number.
(a) Natural (b) Irrational (c) Rational (d) Negative
Ans: (c) Rational

66. Summation or Subtraction of rational number and irrational number is always a / an _____ number.
(a) Natural (b) Whole (c) Rational (d) Irrational
Ans: (d) Irrational

67. By simplifying $(\sqrt{7} + \sqrt{5}) * (\sqrt{7} - \sqrt{5})$, we get _____
 (a) 24 (b) 2 (c) 12 (d) 74

Ans: (b) 2

$$(\sqrt{a} + \sqrt{b}) * (\sqrt{a} - \sqrt{b}) = a - b \Rightarrow (\sqrt{7} + \sqrt{5}) * (\sqrt{7} - \sqrt{5}) = 7 - 5 = 2$$

68. By rationalising the denominator of $1 / \sqrt{3}$, we get _____
 (a) $\sqrt{3} / 3$ (b) $\sqrt{3}$ (c) $3 / \sqrt{3}$ (d) $3 \sqrt{3}$

Ans: (a) $\sqrt{3} / 3$

$$\frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}} * \left(\frac{\sqrt{3}}{\sqrt{3}} \right) = \frac{\sqrt{3}}{3}$$

69. By simplifying $(\sqrt{3} + \sqrt{2}) * (\sqrt{3} + \sqrt{2})$, we get _____
 (a) $5 + 4\sqrt{6}$ (b) $5 + 2\sqrt{6}$ (c) 1 (d) 5

Ans: (b) $5 + 2\sqrt{6}$

$$(\sqrt{a} + \sqrt{b}) * (\sqrt{a} + \sqrt{b}) = a + 2\sqrt{a}\sqrt{b} + b \Rightarrow (\sqrt{3} + \sqrt{2}) * (\sqrt{3} + \sqrt{2}) = 3 + 2\sqrt{3}\sqrt{2} + 2 = 5 + 2\sqrt{6}$$

70. By rationalising the denominator of $1 / (6 - \sqrt{3})$, we get _____
 (a) $(6 - \sqrt{3}) / 3$ (b) $(6 - \sqrt{3}) / 33$ (c) $(6 + \sqrt{3}) / 3$ (d) $(6 + 3\sqrt{3}) / 33$

Ans: (d) $(6 + 3\sqrt{3}) / 33$

$$\frac{1}{(6 - \sqrt{3})} = \frac{1}{(6 - \sqrt{3})} * \frac{6 + \sqrt{3}}{6 + \sqrt{3}} = \frac{6 + \sqrt{3}}{(6 * 6) - (\sqrt{3} * \sqrt{3})} = \frac{6 + \sqrt{3}}{(36 - 3)} = \frac{6 + \sqrt{3}}{33}$$

71. By rationalising the denominator of $1 / (5 + \sqrt{7})$, we get _____
 (a) $(5 - \sqrt{7}) / (\sqrt{5} + \sqrt{7})$ (b) $(5 - \sqrt{7}) / 18$ (c) $(5 + \sqrt{7}) / 18$ (d) $(5 + \sqrt{7}) / (5 - \sqrt{7})$

Ans: (b) $(5 - \sqrt{7}) / 18$

$$\frac{1}{5 + \sqrt{7}} = \frac{1}{5 + \sqrt{7}} * \left(\frac{5 - \sqrt{7}}{5 - \sqrt{7}} \right) = \frac{5 - \sqrt{7}}{(5 * 5) - (\sqrt{7} * \sqrt{7})} = \frac{5 - \sqrt{7}}{25 - 7} = \frac{5 - \sqrt{7}}{18}$$

72. By simplifying $(8)^{2/3}$, we get _____
 (a) 4 (b) 16 (c) $(8)^2 / (8)^3$ (d) 32

Ans: (a) 4

$$8^{2/3} = (8^2)^{1/3} = (64)^{1/3} = \sqrt[3]{64} = 4$$

73. By simplifying $(3)^{3/2}$, we get _____
 (a) $(3)^3 / (3)^2$ (b) $(3)^2 / (3)^3$ (c) $\sqrt{27}$ (d) $9\sqrt{3}$

Ans: (c) $\sqrt{27}$

$$3^{3/2} = (3^3)^{1/2} = 27^{1/2} = \sqrt{27}$$

74. By simplifying $(5)^{3/8}$, we get _____
 (a) $(5)^3 / (5^8)$ (b) $(125)^{1/8}$ (c) $(5)^8 / (5)^3$ (d) 5^{24}

Ans: (b) $(125)^{1/8}$

$$5^{3/8} = (5^3)^{1/8} = (5*5*5)^{1/8} = (125)^{1/8}$$

75. By simplifying $(3)^{1/3} * (3)^{2/3}$, we get _____
 (a) $(3)^{2/9}$ (b) $(3)^{-1/9}$ (c) 9 (d) 3

Ans: (d) 3

76. By simplifying $(7)^{2/5} * (7)^{3/4}$, we get _____
 (a) $(7)^{6/20}$ (b) 7 (c) $(7)^{23/20}$ (d) $(7)^{-7/20}$

Ans: (c) $(7)^{23/20}$

$$a^m a^n = a^{(m+n)} \Rightarrow (7)^{2/5} * (7)^{3/4} = (7)^{(2/5+3/4)} = (7)^{((2*4)+(5*3))/(5*4)} = (7)^{(8+15)} = (7)^{23/20}$$

77. By simplifying $(9)^{1/4} * (9)^{7/4}$, we get _____
 (a) 81 (b) 9 (c) 3 (d) 27

Ans: (a) 81

$$a^m a^n = a^{(m+n)} \quad (9)^{1/4} * (9)^{7/4} = (9)^{(1+7)/4} = (9)^{8/4} = (9)^2 = 81$$

78. By simplifying $(5)^{3/4} / (5)^{1/4}$, we get _____
 (a) $(5)^{1/4}$ (b) $(5)^{3/16}$ (c) $5-\sqrt{3}$ (d) $5-\sqrt{2}$

Ans: (d) $5-\sqrt{2}$

$$a^m / a^n = a^{m-n} \quad (5)^{3/4} / (5)^{1/4} = (5)^{(3/4-1/4)} = (5)^{(3-1)/4} = (5)^{2/4} = (5)^{1/2} = 5-\sqrt{2}$$

79. By simplifying $(4)^9 / (4)^3$, we get _____
 (a) 4^{27} (b) 4^6 (c) 4^{12} (d) 4^3

Ans: (b) 4^6

$$a^m / a^n = a^{m-n} \Rightarrow (4)^9 / (4)^3 = 4^{(9-3)} = 4^6$$

80. By simplifying $(13)^{1/2} / (13)^{7/2}$, we get _____
 (a) $13^{7/4}$ (b) 13^{-3} (c) $113\sqrt{2}$ (d) $113\sqrt{3}$

Ans: (b) 13^{-3}

$$a^m / a^n = a^{m-n} \Rightarrow (13)^{1/2} / (13)^{7/2} = 13^{(1/2-7/2)} = 13^{((1)-(7))/(2)} = 13^{((-6))/(2)} = 13^{-3}$$

81. By simplifying $(4)^{1/3} * (16)^{1/3}$, we get _____
 (a) 4 (b) 8 (c) 64^3 (d) 16

Ans: (a) 4

$$(a^m)*(b^m) = (ab)^m \Rightarrow (4)^{1/3}*(16)^{1/3} = (4*16)^{1/3} = (64)^{1/3} = 4$$

82. By simplifying $2^3 * 5^3$, we get _____
(a) 10^6 (b) 100 (c) 1000 (d) 10^9

Ans: (c) 1000

$$(a^m) * (b^m) = (ab)^m \rightarrow 2^3 * 5^3 = (2*5)^3 = (10)^3 = 1000.$$

83. By simplifying $(2)^{1/2} / (18)^{-1/2}$, we get _____
(a) 36 (b) 36^2 (c) 9 (d) 6

Ans: (d) 6

84. Which of the following is an irrational number?

- (a) 0.75 (b) 2 (c) 4 (d) 0.5

Ans: (b) 2

85. What is the decimal representation of $1 / 3$?

- (a) 0.3 (b) 0.33 (c) 0.333... (Repeating) (d) 0.3333

Ans: (c) 0.333... (Repeating)

86. The number 0.142857142857... is an example of

- (a) Rational number with terminating decimal
(b) Rational number with non-terminating repeating decimal
(c) Irrational number
(d) None of these

Ans: (b) Rational number with non-terminating repeating decimal

87. Which of the following statements is true?

- (a) Every natural number is a whole number
(b) Every whole number is a natural number
(c) Every integer is a whole number
(d) Every rational number is an integer

Ans: (a) Every natural number is a whole number

88. Which of the following is not a rational number?

- (a) $7 / 8$ (b) 0 (c) 3 (d) 0.25

Ans: (c) 3

89. The decimal expansion of $2 / 5$ is

- (a) Terminating (b) Non-terminating repeating
(c) Non-terminating non-repeating (d) None of these

Ans: (a) Terminating

90. Which of the following is an example of an irrational number?

- (a) $3 / 4$ (b) 2.5 (c) π (d) 0.125

Ans: (c) π

91. What is the smallest prime number?

- (a) 0 (b) 1 (c) 2 (d) 3

Ans: (c) 2

92. The number system which includes all rational and irrational numbers is called _____ numbers.

- (a) Natural (b) Whole (c) Real (d) Integer

Ans: (c) Real

93. $(-5 + 2\sqrt{5} - \sqrt{5})$ is a / an _____ number.

- (a) Irrational (b) Positive rational (c) Negative rational (d) Integer

Ans: (a) Irrational

94. $(\sqrt{12} + \sqrt{10} - \sqrt{2})$ is a / an _____ number.

- (a) Positive rational (b) Equal to zero (c) Irrational (d) Negative integer

Ans: (c) Irrational

95. $(-7 + 4\sqrt{7} - 3\sqrt{7})$ is a / an _____ number.

- (a) Positive rational (b) Irrational (c) Negative rational (d) Equal to zero

Ans: (b) Irrational

96. The number $(3 - \sqrt{3})(3 + \sqrt{3})$ is a / an _____ number.

- (a) Irrational (b) Rational (c) Not a natural (d) None of these

Ans: (b) Rational

$$(3 - \sqrt{3})(3 + \sqrt{3}) = (3)^2 - (\sqrt{3})^2 \Rightarrow 9 - 3 = 6$$

97. On simplifying $(\sqrt{5} + \sqrt{7})^2$, we get

- (a) 12 (b) $\sqrt{35}$ (c) $\sqrt{5} + \sqrt{7}$ (d) $12 + 2\sqrt{35}$

Ans: (d) $12 + 2\sqrt{35}$

$$\begin{aligned} (\sqrt{5} + \sqrt{7})^2 &= (\sqrt{5})^2 + 2(\sqrt{5})(\sqrt{7}) + (\sqrt{7})^2 \Rightarrow 5 + 2\sqrt{35} + 7 \\ &= (5 + 7) + 2\sqrt{35} \Rightarrow 12 + 2\sqrt{35} \end{aligned}$$

98. On simplifying $8^3 \times 2^4$, we get

- (a) 16^7 (b) 2^{13} (c) 2^{10} (d) 8^4

Ans: (b) 2^{13}

$$(2^3)^3 \times 2^4 \Rightarrow 2^9 \times 2^4 = 2^{9+4} = 2^{13}$$

99. The value of $\frac{1}{\sqrt{10}}$ when $\sqrt{10} = 3.162$ is

- (a) 0.3162 (b) 31.62 (c) 0.03162 (d) 316.2

Ans: (a) 0.3162

$$\frac{1}{\sqrt{10}} = \frac{1}{3.162} \approx 0.3162555$$

100. $(16)^{3/4}$ is equal to

- (a) 2 (b) 4 (c) 8 (d) 16

Ans: (c) 8

$$(2^4)^{3/4} = 2^{4 \cdot \frac{3}{4}} = 2^3 = 8$$